

DATE: EXPT NO:03	EDA-Data Inspection and Analysis
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AIM:

To inspect and analyze the Research Student dataset using Exploratory Data Analysis (EDA) to understand the data structure, identify patterns, detect missing values, and discover relationships between different student attributes.

PROCEDURE:

1. Import Pandas, NumPy, Matplotlib, and Seaborn libraries.
2. Load the Research Student dataset using Pandas.
3. Display the first and last records using head() and tail().
4. Check the number of rows and columns using shape.
5. Display column names and data types using info().
6. Check missing values using isnull().sum().
7. Generate statistical summaries using describe().
8. Check and identify duplicate records.
9. Analyze important student-related attributes.
10. Create suitable graphs such as histograms, bar charts, and scatter plots.
11. Generate a correlation heatmap to identify relationships between numerical variables.
12. Interpret the findings and identify important patterns in the Research Student dataset.

CODE:

```
import pandas as pd  
  
import numpy as np  
  
import matplotlib.pyplot as plt  
  
import seaborn as sns
```

```
file_path = r"C:\Users\babyn\Downloads\archive\research_student (1).xlsx"
```

```
df = pd.read_excel(file_path)
```

```
print("Dataset Preview:")
```

```
print(df.head())
```

```
# 1. Identifying Missing Values
```

```
# -----
```

```
print("\nMissing Values in Each Column:")
```

```
print(df.isnull())
```

```
# -----
```

```
# 2. Count Missing Values
```

```
# -----
```

```
print("\nTotal Missing Values:")
```

```
print(df.isnull().sum())
```

```
# Import required libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.preprocessing import MinMaxScaler
```

```
# -----
```

```
# Load Dataset
```

```
# -----
```

```
file_path = r"C:\Users\babyn\Downloads\archive\research_student (1).xlsx"
```

```
df = pd.read_excel(file_path)
```

```
print("Original Dataset:")
```

```
print(df.head())
```

```
print("\nDataset Information:")
```

```
print(df.info())
```

```
# -----
```

```
# 1. Identify Missing Values
```

```
# -----
```

```
print("\nMissing Values in Dataset:")
```

```
print(df.isnull())
```

```
# -----
```

```
# 2. Count Missing Values
```

```
# -----
```

```
print("\nTotal Missing Values in Each Column:")
print(df.isnull().sum())

# -----

# 3. Replace Missing Values
# -----

# Numerical columns - replace missing values with mean

numeric_columns = df.select_dtypes(include=np.number).columns

for col in numeric_columns:
    df[col] = df[col].fillna(df[col].mean())

# Categorical columns - replace missing values with mode

categorical_columns = df.select_dtypes(include="object").columns

for col in categorical_columns:
    df[col] = df[col].fillna(df[col].mode()[0])

print("\nMissing Values After Replacement:")
print(df.isnull().sum())

# -----

# 4. Detect Duplicate Records
```

```
# -----
```

```
print("\nNumber of Duplicate Rows:")
```

```
print(df.duplicated().sum())
```

```
# Remove duplicate records
```

```
df = df.drop_duplicates()
```

```
print("\nDuplicate Rows After Removing:")
```

```
print(df.duplicated().sum())
```

```
# -----
```

```
# 5. Detect Outliers using IQR Method
```

```
# -----
```

```
print("\nOutlier Detection:")
```

```
outlier_result = {}
```

```
for col in numeric_columns:
```

```
    Q1 = df[col].quantile(0.25)
```

```
    Q3 = df[col].quantile(0.75)
```

```
    IQR = Q3 - Q1
```

```
lower_bound = Q1 - (1.5 * IQR)
```

```
upper_bound = Q3 + (1.5 * IQR)
```

```
outliers = df[
```

```
    (df[col] < lower_bound) |
```

```
    (df[col] > upper_bound)
```

```
]
```

```
outlier_result[col] = len(outliers)
```

```
print(outlier_result)
```

```
# -----
```

```
# 6. Outlier Visualization
```

```
# -----
```

```
for col in numeric_columns:
```

```
    plt.figure(figsize=(6,4))
```

```
    sns.boxplot(x=df[col])
```

```
    plt.title("Outlier Detection : " + col)
```

```
plt.show()
```

```
# -----
```

```
# 7. Data Cleaning
```

```
# -----
```

```
# Remove spaces from column names
```

```
df.columns = df.columns.str.strip()
```

```
# Convert column names into lowercase
```

```
df.columns = df.columns.str.lower()
```

```
print("\nCleaned Column Names:")
```

```
print(df.columns)
```

```
# -----
```

```
# 8. Data Transformation
```

```
# -----
```

```
# Normalization using MinMaxScaler
```

```
scaler = MinMaxScaler()
```

```
df[numeric_columns]
```

```
print("\nTransformed Dataset:")
```

```
output_file = "cleaned_research_student.xlsx"
```

```
df.to_excel(
```

```
print("\nCleaned Dataset Saved Successfully!")
```

OUTPUT:

Original Dataset:

Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category \
--------	-------------	-------------	--------	-------------	-------------	------------

0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
---	-----	-----	-----	-----	-----	-----	-----

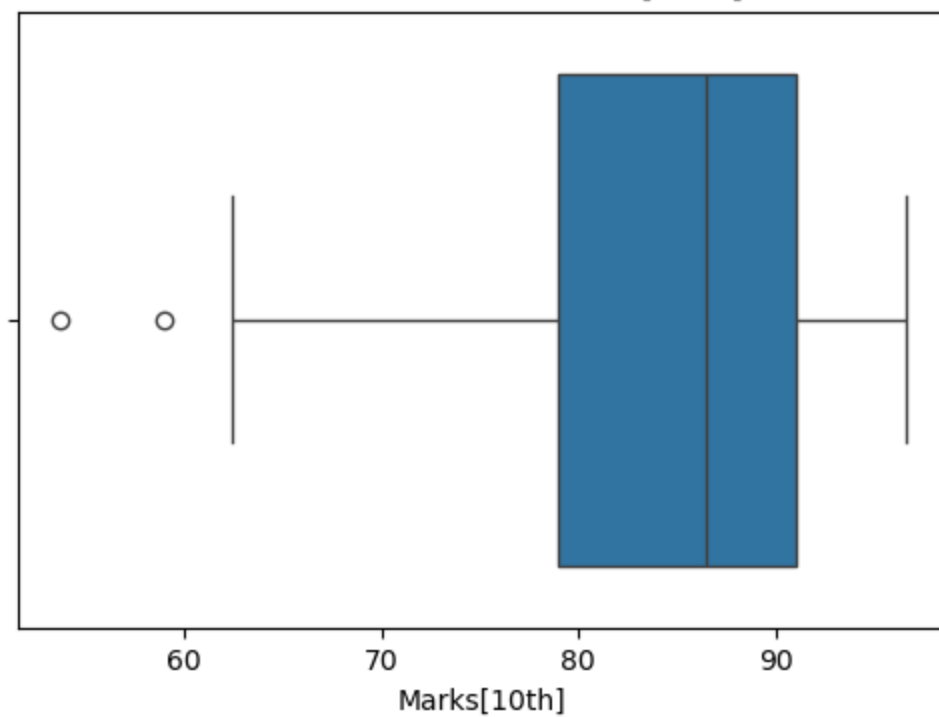
1	CIVIL	77.57	64.6	Male	BSEB Patna	BSEB Patna	OBC
2	CSE	86.40	71.8	Male	CBSE	CBSE	GEN
3	CSE	88.14	78.0	Male	ICSE	ICSE	GEN
4	CSE	65.40	59.8	Female	CBSE	CBSE	ST

	GPA 1	Rank	Normalized Rank ...	GPA 3	GPA 4	GPA 5	GPA 6 \
0	NaN	NaN	NaN ...	NaN	NaN	NaN	NaN
1	6.29	44718.0	15.970714 ...	5.94	5.41	6.25	6.13
2	6.47	24222.0	8.650714 ...	5.88	5.53	6.44	6.19
3	7.35	24723.0	8.829643 ...	6.54	6.41	6.50	6.69
4	6.41	232157.0	82.913214 ...	5.71	5.24	5.88	6.25

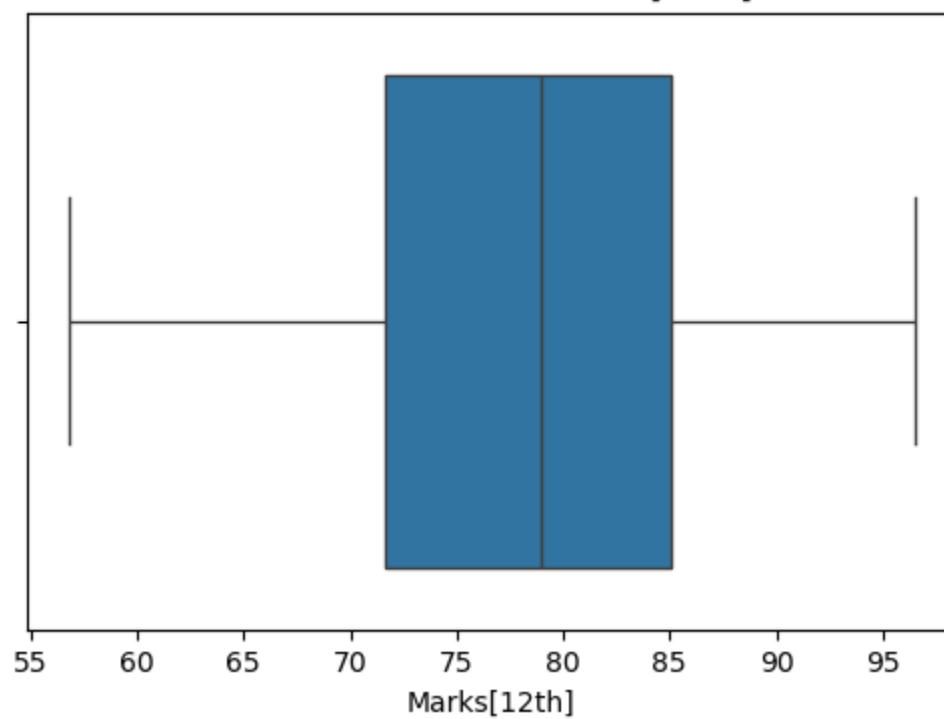
	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching \
0	NaN	NaN	NaN	NaN
1	1.0	4.0	3.0	2.0
2	2.0	2.0	0.0	1.0
3	1.0	1.0	3.0	0.0
4	2.0	0.0	2.0	2.0

	NTSE Scholarships	Miscellany Tech Events
0	NaN	NaN
1	0.0	5.0
...		

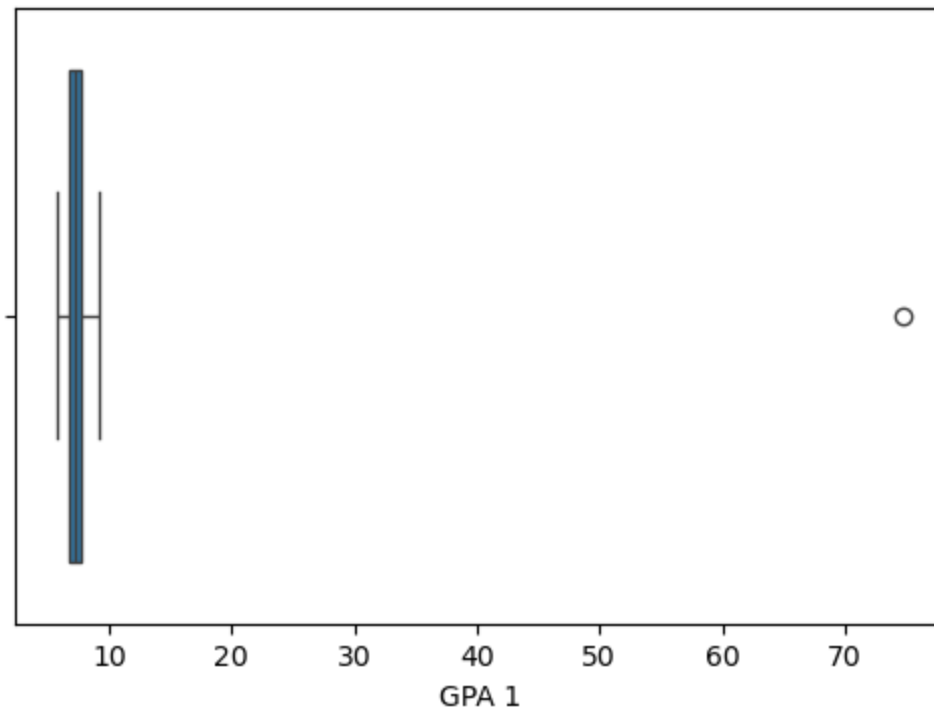
Outlier Detection : Marks[10th]



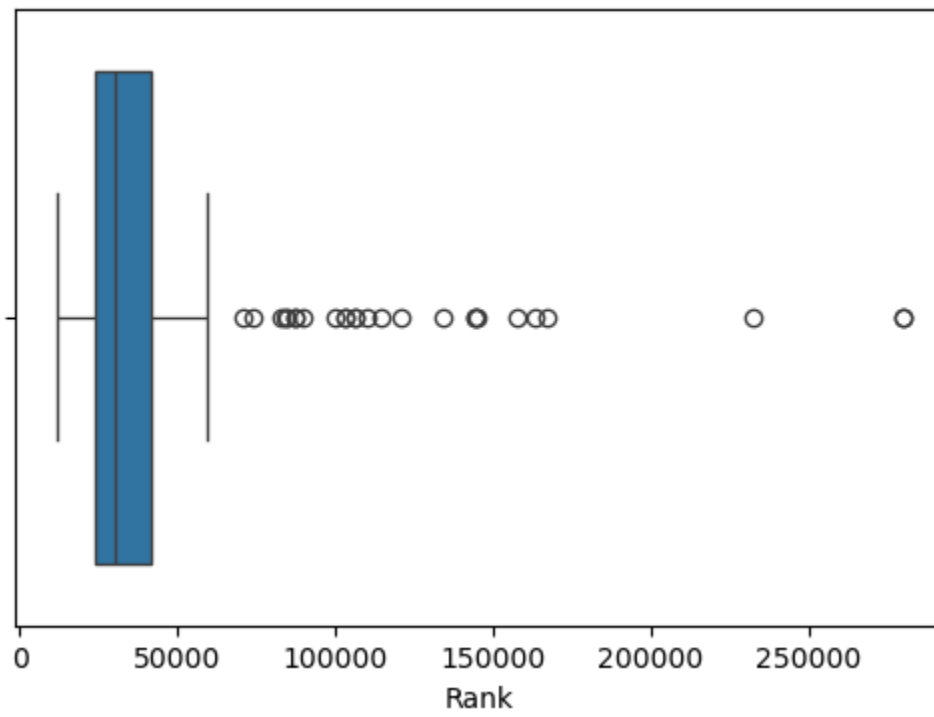
Outlier Detection : Marks[12th]

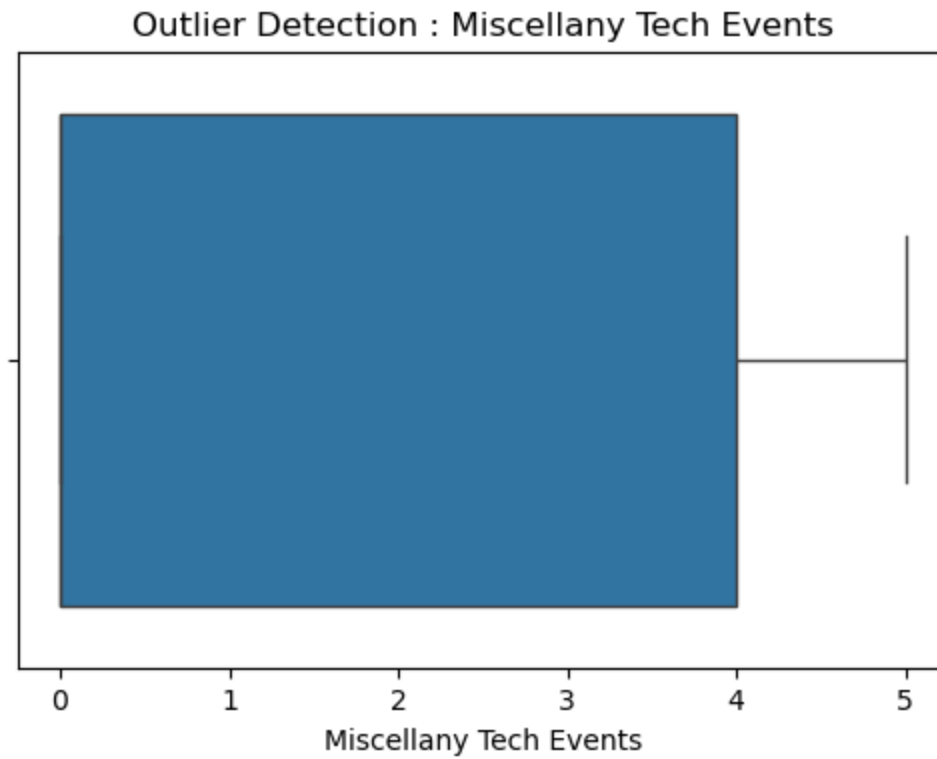


Outlier Detection : GPA 1



Outlier Detection : Rank





RESULT:

The Research Student dataset was successfully inspected and analyzed using EDA. The dataset structure, missing values, duplicates, statistical details, and important relationships were identified through data analysis and visualization.